BACKBONE V2.1
Medulla Oblongata

1	24V	FIELD 24 V
2	24V	Paralelné napájanie
3	GND	FIELD GND
4	GND	FIELD GND
5	5V	FIELD 5 V
6	5V	Paralelné napájanie
7	NC	Rezerva
8	NC	Rezerva
9	24V_LOGIC	LOGIC 24 V
10	24V_LOGIC	Paralelné napájanie
11	LOGIC_GND	Logická zem
12	LOGIC_GND	Logická zem
13	NC	Rezerva
14	NC	Rezerva
15	24V_BATT	Batériová vetva 24 V
16	5V_BATT	Batériová vetva 5 V
17	BATT_GND	Batériová zem
18	BATT_GND	Batériová zem
19	5V_LOGIC	LOGIC 5 V
20	LOGIC_GND	Logická zem
21	NC	Rezerva
22	NC	Rezerva
23	I2C_SCL	I²C hodinový signál
24	I2C_SDA	I²C dátový signál
25	NC	Rezerva pre budúce IRQ
26	NC	Rezerva
27	SPI_SCK	SPI clock
28	LOGIC_GND	Referenčná zem SPI
29	SPI_MOSI	SPI controller → periféria
30	SPI_MISO	SPI periféria → controller
31	SPI_CS0	Výber SPI zariadenia 0
32	SPI_CS1	Výber SPI zariadenia 1
33	SPI_DC	Display Data/Command
34	LOGIC_GND	Referenčná zem SPI
35	NC	Rezerva
36	RS485_GND	Referenčná zem RS-485
37	RS485_A	Diferenciálna linka A
38	RS485_B	Diferenciálna linka B
39	NC	Rezerva
40	NC	Rezerva

H101
M20-6102045

R120
MBB02070Z0000ZC100

MANUS

IN LINE PULL UP ULNA

R102 ERJPA3F4701V
R103 ERJ-3EKF1001V
C102 C0603C103K5RACTU
R104 ERJPA3F4701V
R105 ERJ-3EKF1001V
C103 C0603C103K5RACTU
R106 ERJPA3F4701V
R107 ERJ-3EKF1001V
C104 C0603C103K5RACTU
R108 ERJPA3F4701V
R109 ERJ-3EKF1001V
C105 C0603C103K5RACTU

LOGIC_3V3, INDEX, POLLEX, DIGITUS_MEDIUS, DIGITUS_ANULARIS, LOGIC_GND

CN101
1990067

DIGITUS MINIMUS

0-30V OUT LINE RADIUS

Pull-down na GPIO
R112 ERJ2RKF4702X
Obmedzenie prúdu LED
R110 ERJ3EKF3300V
U101 AQY211EH
OS_Trapenzium
D101 SMBJ26CA_C83332

Pull-down na GPIO
R111 ERJ2RKF4702X
Obmedzenie prúdu LED
R113 ERJ3EKF3300V
U102 AQY211EH
OS_Trapezoideum
D102 SMBJ26CA_C83332

Pull-down na GPIO
R114 ERJ2RKF4702X
Obmedzenie prúdu LED
R115 ERJ3EKF3300V
U104 AQY211EH
OS_Capitatum
D103 SMBJ26CA_C83332

Pull-down na GPIO
R116 ERJ2RKF4702X
Obmedzenie prúdu LED
R117 ERJ3EKF3300V
U105 AQY211EH
OS_Hamatum
D104 SMBJ26CA_C83332

CN102
1990067

ESP32-S3-WROOM-1U-N16R8

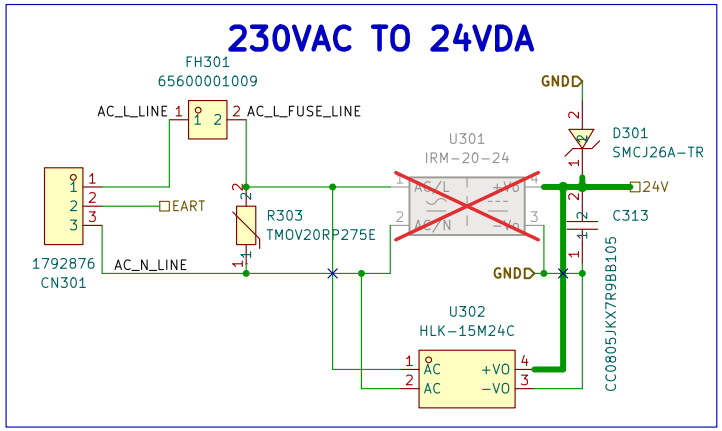
POLLEX, INDEX, DIGITUS_MEDIUS, DIGITUS_ANULARIS, OS_Trapenzium, OS_Trapezoideum, OS_Capitatum, OS_Hamatum, RS485_A, RS485_B, LOGIC_GND, LOGIC_3V3, I2C_SCL, I2C_SDA, JTAG TDI, JTAG TMS, RGB_DATA, SPI_CS1, SPI_CS2, SPI_MISO, SPI_MOSI, SPI_SCK, SPI_DC, UART0_RX, UART0_TX, UART_RX, UART_TX, CHASIS

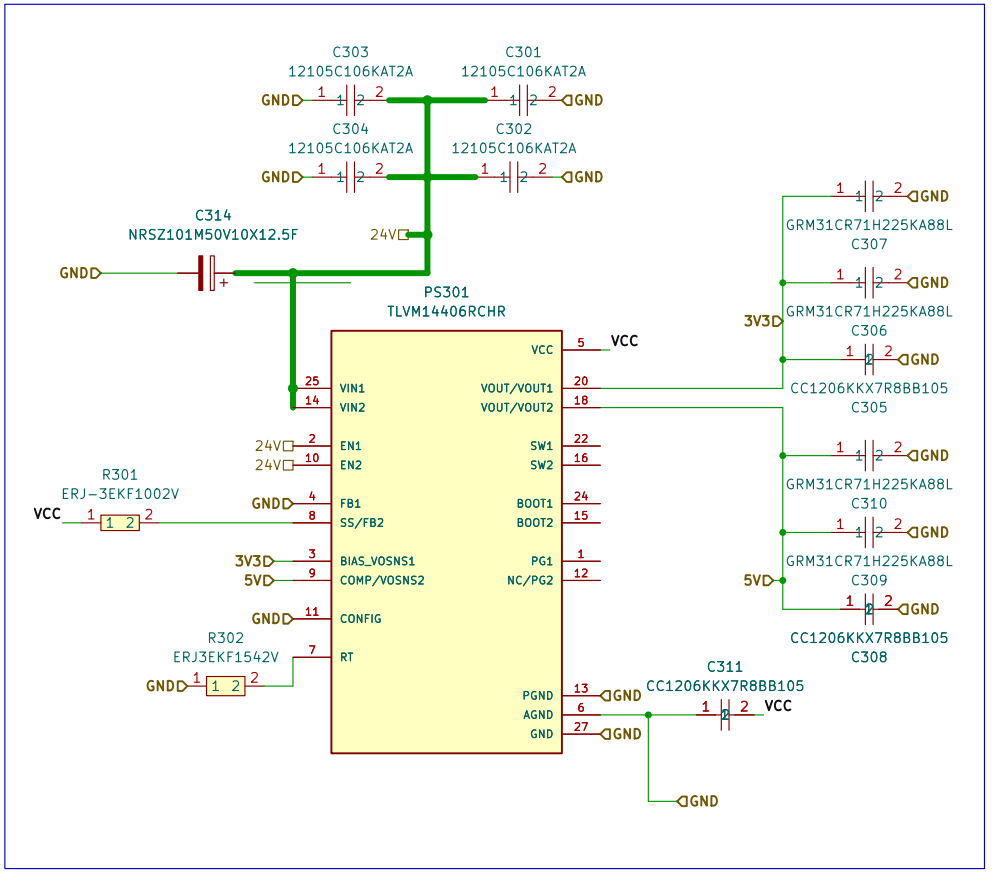
R121 MBB02070Z0000ZC100
R119 MBB02070Z0000ZC100
C107 DE1E3RA102MA4BN01F
R122 CRCW12061M00FKEA
36V_TO_5V+3V3
CHASSIS_GND

RGB

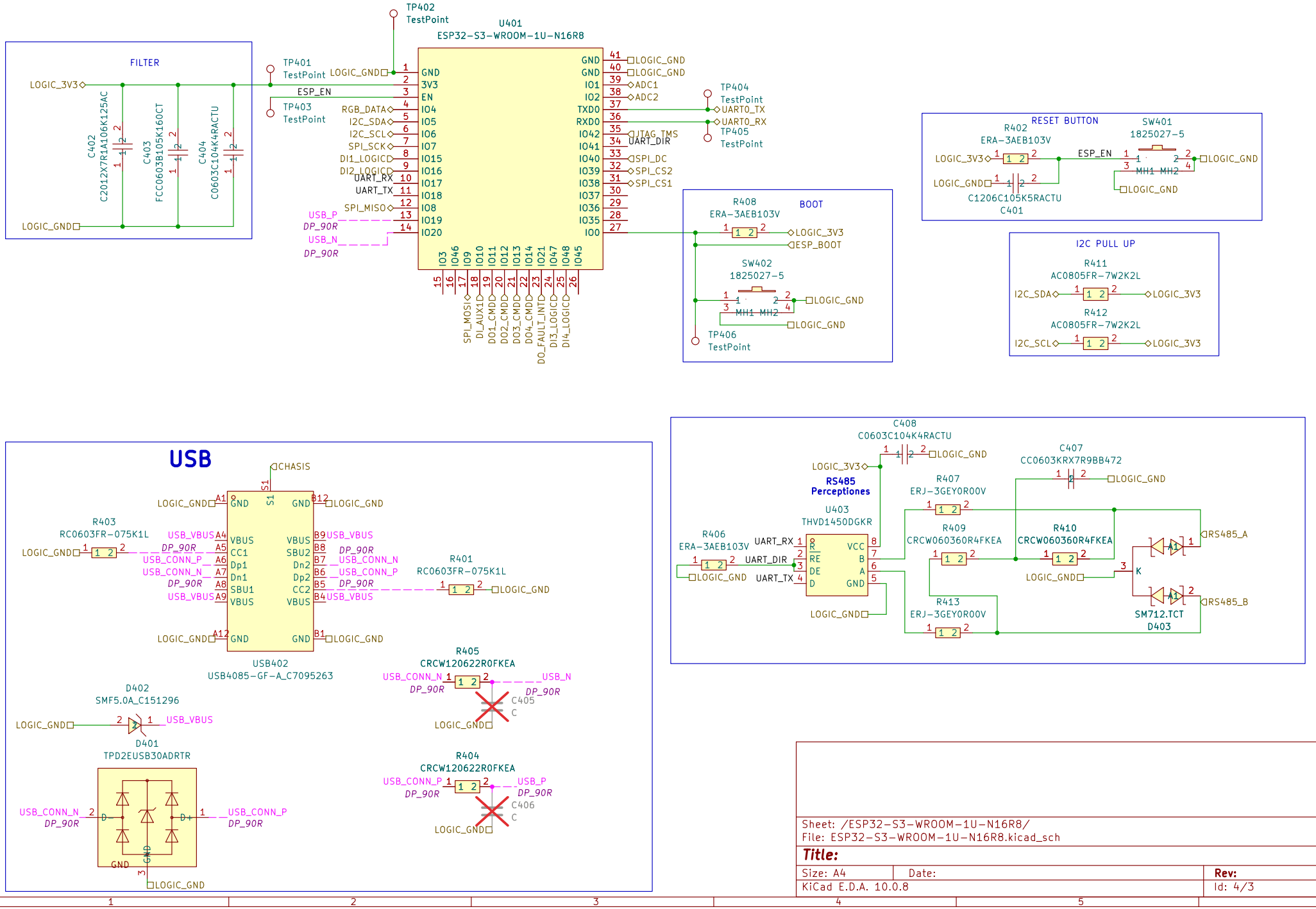
U103 SN74AHCT1G125DBVR
C101 C0805C104K5RACTU_C3018562
R118 ERA-3AEB103V
R101 ERJ3EKF3300V
CN104 1989751

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Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 1/3





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File: 36v to 5v + 3v3.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 3/3


FILTER
C402
C2012X7R1A106K125AC
C403
FCC0603B105K160CT
C404
C0603C104K4RACTU
LOGIC_3V3
LOGIC_GND
TP402
TestPoint
U401
ESP32-S3-WROOM-1U-N16R8
TP401
TestPoint
TP403
TestPoint
ESP_EN
RGB_DATA
I2C_SDA
I2C_SCL
SPI_SCK
DI1_LOGICD
DI2_LOGICD
UART_RX
UART_TX
SPI_MISO
USB_P
DP_90R
USB_N
DP_90R
TP404
TestPoint
UART0_TX
UART0_RX
TP405
TestPoint
JTAG_TMS
UART_DIR
SPI_DC
SPI_CS2
SPI_CS1
ADC1
ADC2
R408
ERA-3AEB103V
BOOT
LOGIC_3V3
ESP_BOOT
SW402
1825027-5
LOGIC_GND
TP406
TestPoint
RESET BUTTON
R402
ERA-3AEB103V
SW401
1825027-5
C1206C105K5RACTU
C401
I2C PULL UP
R411
AC0805FR-7W2K2L
R412
AC0805FR-7W2K2L
USB
USB402
USB4085-GF-A_C7095263
R403
RC0603FR-075K1L
R401
RC0603FR-075K1L
R405
CRCW120622R0FKEA
C405
R404
CRCW120622R0FKEA
C406
D402
SMF5.0A_C151296
D401
TPD2EUSB30ADRTR
C408
C0603C104K4RACTU
C407
CC0603KRX7R9BB472
RS485
Perceptiones
U403
THVD1450DGKR
R406
ERA-3AEB103V
R407
ERJ-3GEY0R00V
R409
CRCW060360R4FKEA
R410
CRCW060360R4FKEA
R413
ERJ-3GEY0R00V
SM712.TCT
D403
RS485_A
RS485_B
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Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.8
Id: 4/3